

**Department of Science and Humanities**

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|---------------------|---------------------------------------------------------------------------------|------------------|------------|
| <b>Course Name:</b> | <b>Basics Electrical Engineering(BEE)</b><br><br><b>Course Code: 316U06C104</b> | <b>Semester:</b> | <b>I</b>   |
| <b>Exam:</b>        | <b>BEE IA1(Simulation based Assignment )</b>                                    | <b>Division:</b> | <b>All</b> |

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**Div-Batch:** F-1

**Course:** BEE IA-1

**Q1.** Calculate the current through  $R1 \Omega$  resistor using Mesh analysis. In the circuit shown in figure 1, the value of  $R1$  will be the **last two digits of roll no. times  $\Omega$**

For e.g.: For Roll no: 1000020, then  $R1 = 20\Omega$ . If your last digits are from 01 to 10, kindly add 10 to your last two digits, so roll no with last two digits with 01 will become 11, then  $R1 = 11\Omega$  **(ODD ROLL NO ATTEMPT Q1)**

- Simulate the circuit shown in figure 1, using LT spice software and measure  $I_{R1}$ ,  $I_{4\Omega}$  and  $I_{7\Omega}$  and  $I_{3\Omega}$ .
- Also solve the numerical theoretically and calculate  $I_{R1}$ ,  $I_{4\Omega}$  and  $I_{7\Omega}$  and  $I_{3\Omega}$ .

Tabulate the results as shown in the table below.

| Parameter     | Theoretical value | Simulated value |
|---------------|-------------------|-----------------|
| $I_{R1}$      | 0.286A            | 0.286A          |
| $I_{4\Omega}$ | 3.468A            | 3.468A          |
| $I_{7\Omega}$ | 0.516A            | 0.516A          |
| $I_{3\Omega}$ | 3.238A            | 3.238A          |

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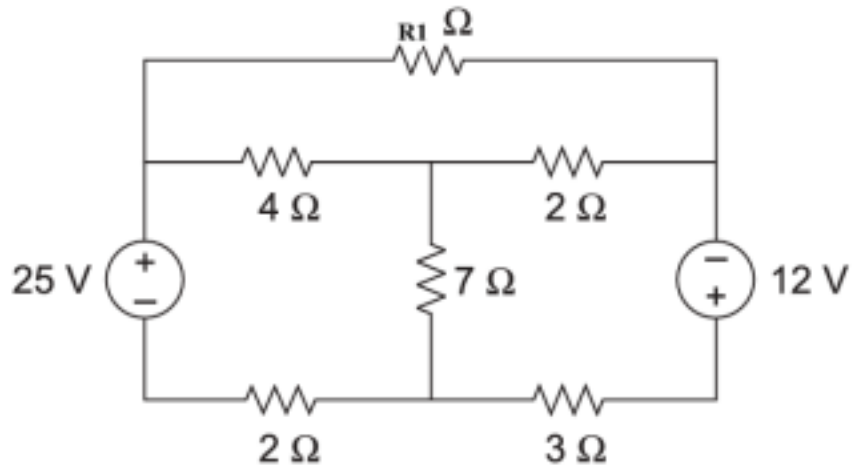
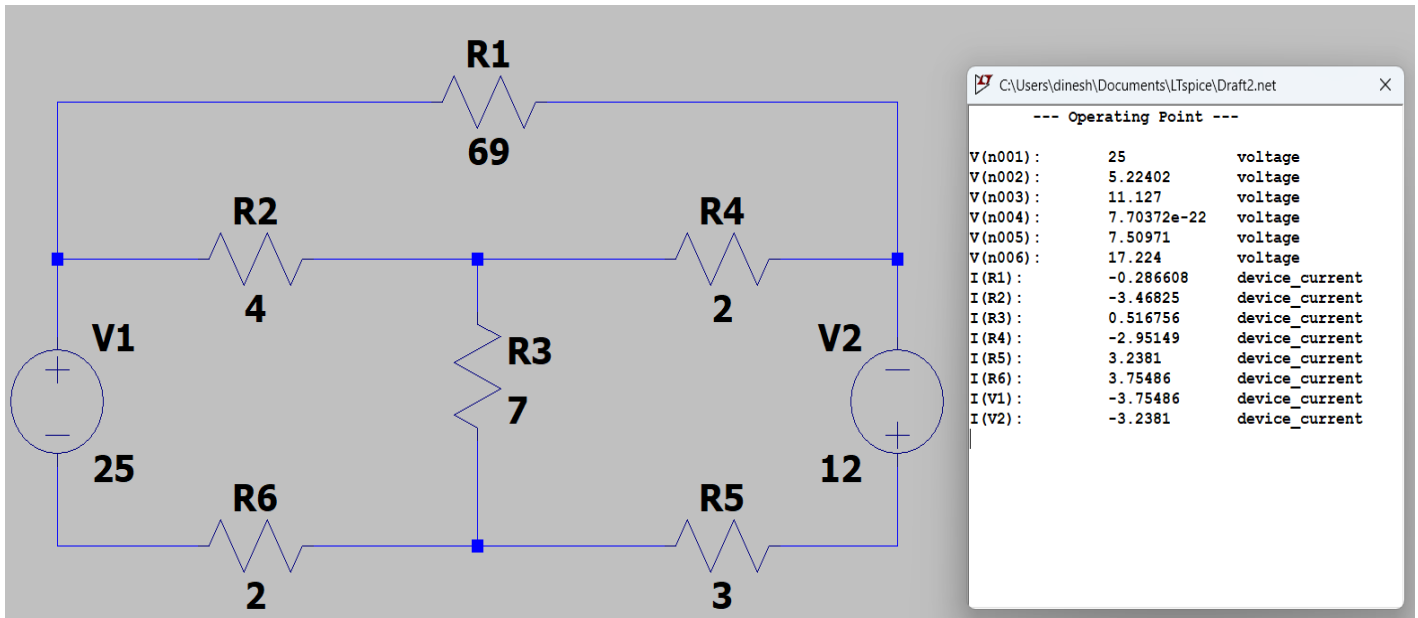
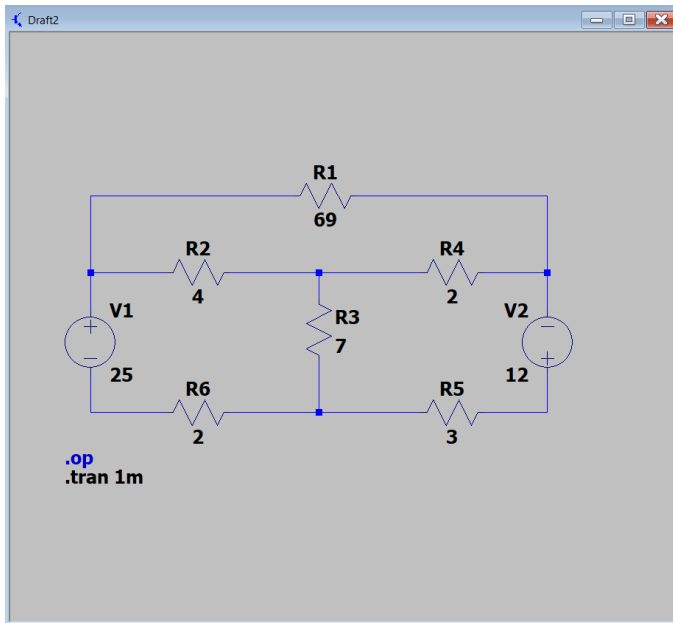


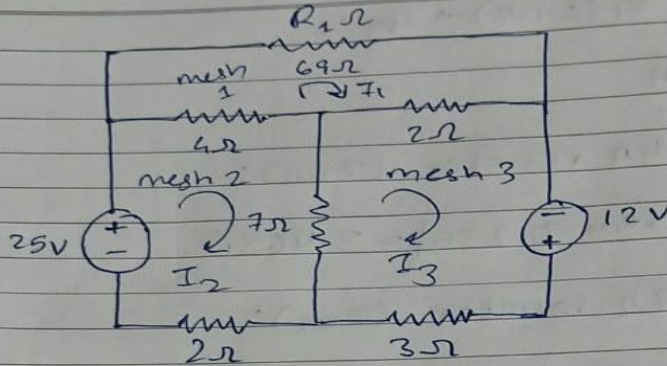
Figure 1







Q1



$$R_1 = 69\Omega$$

→ Applying KVL in mesh 1:

$$(R_1 + 4)I_1 - 4I_2 - 2I_3 = 0$$

$$(69 + 4)I_1 - 4I_2 - 2I_3 = 0$$

$$73I_1 - 4I_2 - 2I_3 = 0 \quad \dots (i)$$

Applying KVL in mesh 2:

$$-25 + 4(I_2 - I_1) + 7(I_2 - I_3) + 2I_2 = 0$$

$$-4I_1 - 7I_3 + (4 + 7 + 2)I_2 = 25$$

$$-4I_1 + 13I_2 - 7I_3 = 25 \quad \dots (ii)$$

Applying KVL in mesh 3:

$$7(I_3 - I_2) + 2(I_3 - I_1) - 12 + 3I_3 = 0$$

$$-2I_1 - 7I_2 + (7 + 2 + 3)I_3 = 12$$

$$-2I_1 - 7I_2 + 12I_3 = 12 \quad \dots (iii)$$

Solving (i), (ii) and (iii) we get,

$$I_1 = 0.286A$$

$$I_2 = 3.754A$$

$$I_3 = 3.238A$$

Current through  $R_1 (69\Omega) \rightarrow I_{R_1}$ :

$$I_{R_1} = I_1 = 0.286A$$

Current through  $4\Omega$  resistor ( $I_{4\Omega}$ ):

$$I_{4\Omega} = |I_2 - I_1| = |3.754 - 0.286| = 3.468A$$

Current through  $7\Omega$  resistor ( $I_{7\Omega}$ ):

$$I_{7\Omega} = |I_2 - I_3| = |3.754 - 3.238| = 0.516A$$

Current through  $3\Omega$  resistor ( $I_{3\Omega}$ ):

$$I_{3\Omega} = I_3 = 3.238A$$